Electrostatic Boundary Conditions

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Electrostatic Boundary Conditions

- If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called boundary conditions.

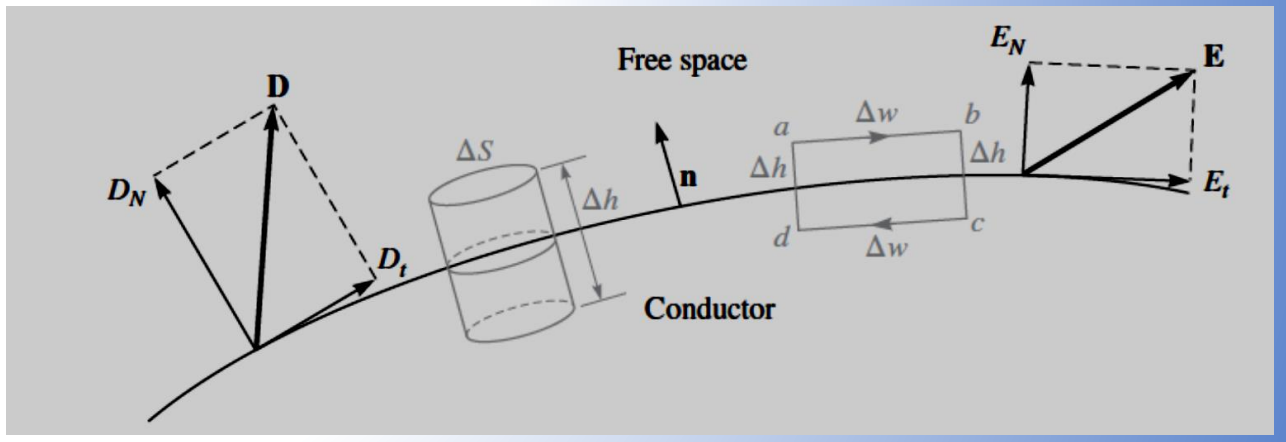
- Conductor and Free space

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$E_t \Delta w - E_{N,\text{at } b} \frac{1}{2} \Delta h + E_{N,\text{at } a} \frac{1}{2} \Delta h = 0$$

$$E_t = 0.$$

As Δh tends to zero



$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q$$

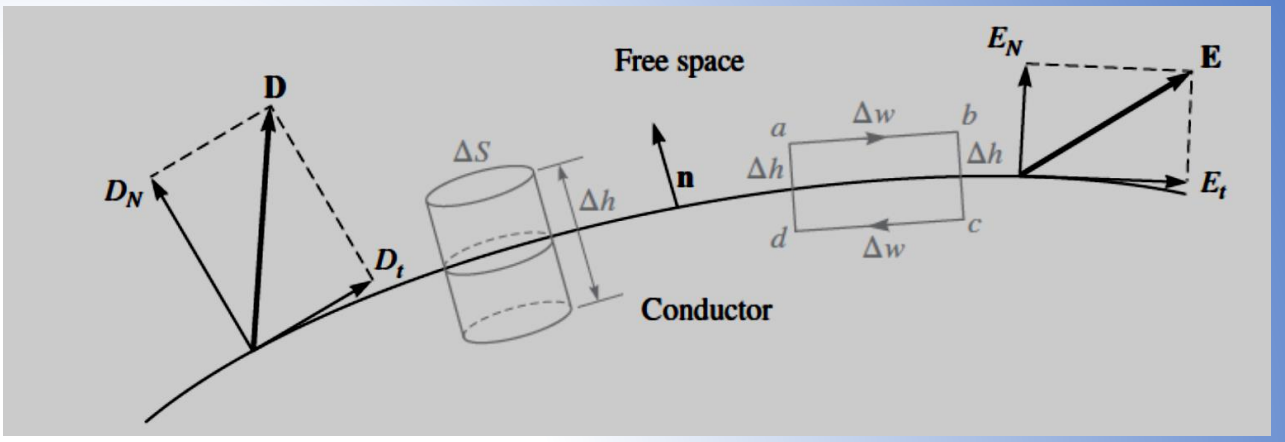
$$\int_{\text{top}} + \int_{\text{bottom}} + \int_{\text{sides}} = Q$$

$$D_N \Delta S = Q = \rho_S \Delta S$$

$$D_N = \rho_S$$

$$D_t = E_t = 0$$

$$D_N = \epsilon_0 E_N = \rho_S$$



1. The static electric field intensity inside a conductor is zero.
2. The static electric field intensity at the surface of a conductor is everywhere directed normal to that surface.
3. The conductor surface is an equipotential surface.

BOUNDARY CONDITIONS FOR PERFECT DIELECTRIC MATERIALS

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$E_{\tan 1}\,\Delta w - E_{\tan 2}\,\Delta w = 0$$

$$\boxed{E_{\tan 1} = E_{\tan 2}}$$

$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\epsilon_1}{\epsilon_2}$$

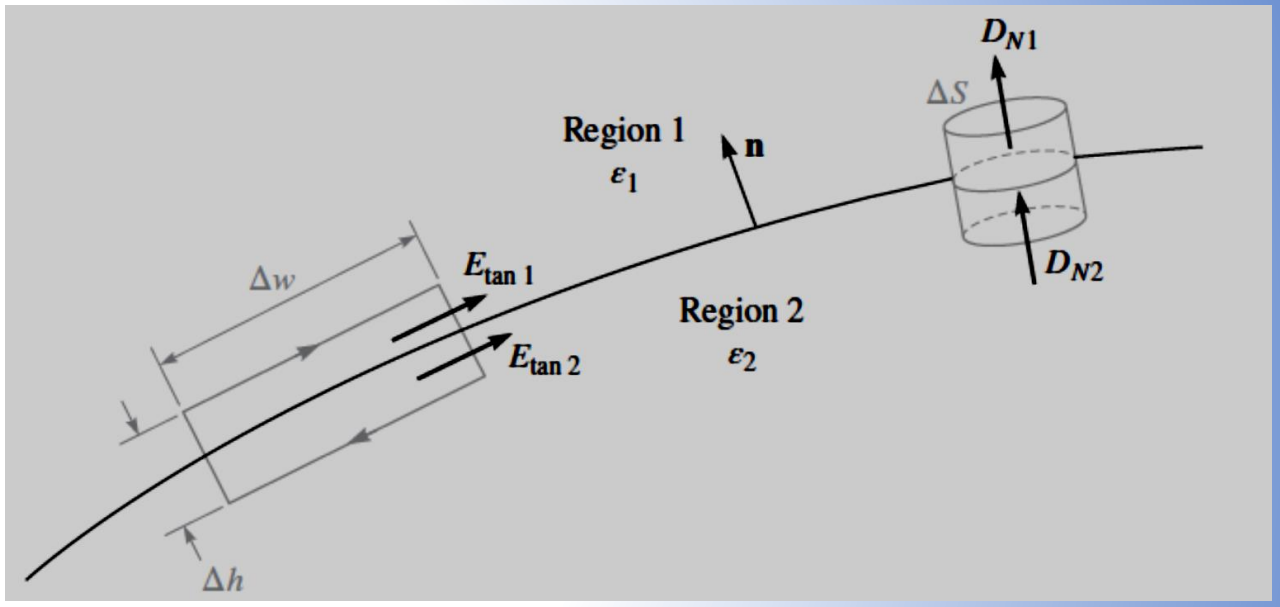
$$D_{N1}\,\Delta S - D_{N2}\,\Delta S = \Delta Q = \rho_S\,\Delta S$$

$$\boxed{D_{N1} - D_{N2} = \rho_S}$$

If $\rho_s = 0$

$$\boxed{D_{N1} = D_{N2}}$$

$$\epsilon_1 E_{N1} = \epsilon_2 E_{N2}$$



$$D_{N1} = D_{N2}$$

$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\epsilon_1}{\epsilon_2}$$

$$D_{N1} = D_1 \cos\theta_1 = D_2 \cos\theta_2 = D_{N2}$$

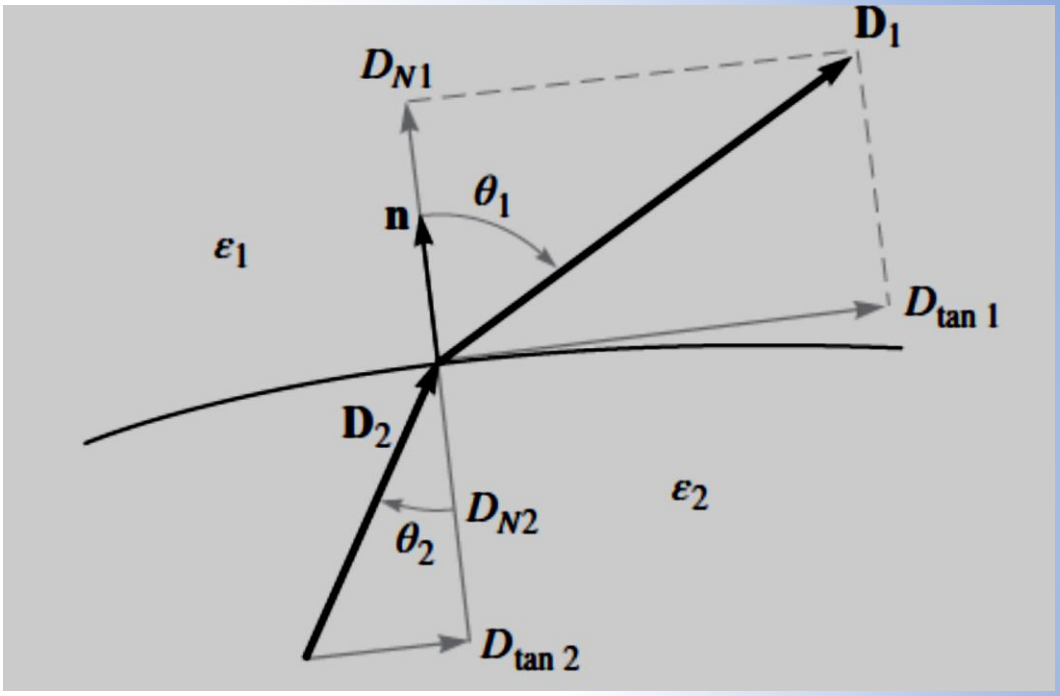
$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{D_1 \sin\theta_1}{D_2 \sin\theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

$$\epsilon_2 D_1 \sin\theta_1 = \epsilon_1 D_2 \sin\theta_2$$

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

$$D_2 = D_1 \sqrt{\cos^2\theta_1 + \left(\frac{\epsilon_2}{\epsilon_1}\right)^2 \sin^2\theta_1}$$

$$E_2 = E_1 \sqrt{\sin^2\theta_1 + \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 \cos^2\theta_1}$$




Thank
you